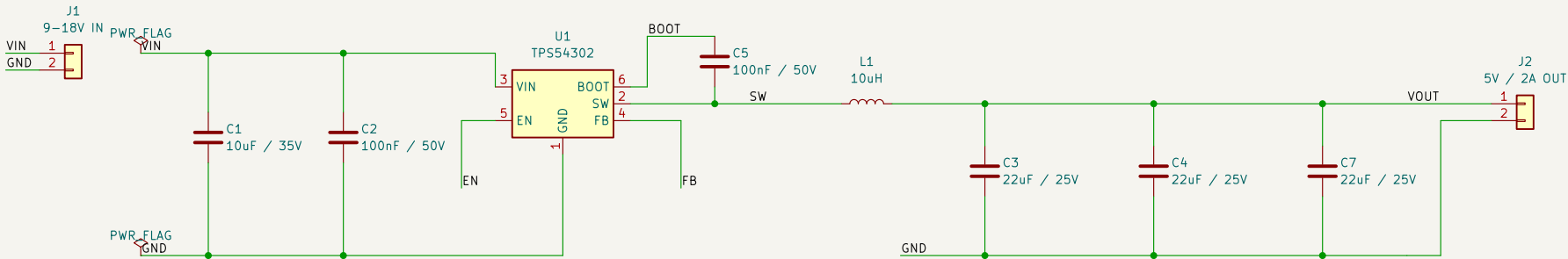


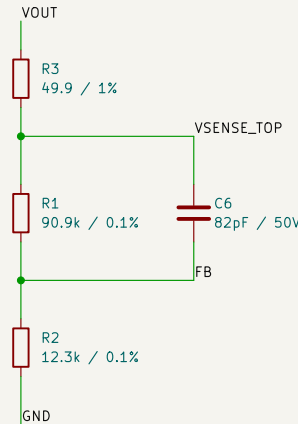
POWER STAGE | 10 W DESIGN TARGET



INPUT ENABLE / UVLO



OUTPUT FEEDBACK / FEED-FORWARD



R3 is the EVM series loop-injection resistor.
R1 and C6 are in parallel, downstream of R3.
Divider resistors: 0.1%; nominal VOUT = 5.003 V.
Capacitor effective values require DC-bias review.

Two-layer PCB; 54 x 34 mm; 1.6 mm FR-4.
Regulator pin mapping checked against TI datasheet.
No reverse polarity or surge protection provided.
Use a current-limited 9-18 V laboratory source.

TPS54302; adapted from TI SLVSDG6C Fig. 7-1 / SLVUAP9B EVM

Engineering portfolio | Designed and simulated – not hardware tested

Sheet: /

File: circuit-design.kicad_sch

Title: 9-18V to 5V / 2A synchronous buck

Size: A4

Date: 2026-10-10

Rev: A

KiCad E.D.A. 10.0.6

Id: 1/1